Clinical Oncology 29 (2017) 562–567

Contents lists available at ScienceDirect

Clinical Oncology

journal homepage: www.clinicaloncologyonline.net

Original Article

Incident Learning Systems for Radiation Oncology: Development and Value at the Local, National and International Level

T. Pawlicki [*], M. Coffey [†], M. Milosevic [‡§]

[*] Department of Radiation Medicine and Applied Sciences, University of California San Diego, La Jolla, California, USA
[†] Discipline of Radiation Therapy, School of Medicine, Trinity College, Dublin, Ireland
[‡] University Health Network and Princess Margaret Cancer Center, Toronto, Canada
[§] University of Toronto, Toronto, Canada

Received 19 May 2017; received in revised form 6 July 2017; accepted 7 July 2017

Abstract

Aims: To discuss the background for incident reporting and learning systems, as well as the infrastructure and operational aspects to run them.
Materials and methods: Information from peer-reviewed literature, online resources and the authors' experience synthesised into a concise understanding of the topic.
Results: Incident learning systems can be local, national or international, each having the same basic goals but facilitating different audiences and environments. A key component of any reporting and learning system is timely and effective analysis of near-misses and incidents as well as feedback to the users of the system. It is important for staff to know that reports are acknowledged, analysed and acted upon. There is a need to comply with current European legislation and other national systems, which can be addressed together with the steps required for comprehensive management of an incident.
Conclusion: Reporting and learning from incidents and near-misses is a key component of quality and safety in radiotherapy. A major benefit of the national or international systems is the potential for a larger database of incidents, supporting wider analysis and comparison, and sharing of knowledge across a larger community.

Key words: Causal analysis; incident learning system; reporting

Introduction

Radiotherapy is generally considered to be safe and has a long history of dosimetric monitoring and, more recently, clinical audit. However, given the level of complexity and the number of different professionals involved, the potential for error and the consequences of an error can be catastrophic [1]. Systematic errors can affect many patients, whereas random errors usually affect one or a few patients. In either case, incident learning systems (ILSs) have been shown to be valuable in elucidating these sources of error [2].

This paper is aimed at departments, individual government regulatory bodies and groups of countries that have recently implemented or are considering implementing an ILS for radiation oncology. As incidents involving unintended exposure to staff and the general public typically require mandatory reporting to the relevant national radiation safety regulatory body, the focus of this work is primarily on patient-related incidents and near-misses. An 'actual incident' refers to an error that affected the patient and is of any severity. A 'near-miss' is an incident that did not reach the patient, possibly because it was caught by an existing safety barrier. In this work, the term 'event' is used to denote both actual incidents and near-misses.

An ILS is one tool in the broader context of incident identification, response, investigation and learning. Reporting an incident, in and of itself, is of little value if incidents are not investigated and analysed appropriately. The goals of any ILS are to enable learning for the purpose of minimising the probability or severity of future incidents, to inform an investigation into the causative factors and to

Author for correspondence: T. Pawlicki, Department of Radiation Medicine and Applied Sciences, University of California San Diego, La Jolla, California, USA.

E-mail address: tpaw@ucsd.edu (T. Pawlicki).

http://dx.doi.org/10.1016/j.clon.2017.07.009
0936-6555/

support an appropriate response. The name 'incident learning system' is somewhat of a misnomer in that one also hopes to capture and learn from near-misses. Furthermore, learning from information within an ILS takes place outside the system and requires an appropriate safety culture. The focus on learning is also emphasised by the Institute of Medicine, who stated that it was imperative that all health care providers develop comprehensive patient safety systems that promote learning [3], and in the European Council Directive 201/-59/EURATOM [4], which makes reporting and learning from incidents a legal requirement.

Learning from events raises awareness within a community but to achieve the maximum impact and outcome it must be managed carefully. Reporting can only take place in a secure environment where the threat of repercussion has been removed. A hierarchical management structure in some settings can mitigate against reporting for fear of sanctions. Blaming individuals or organisations is a factor in the failure to improve safety. Therefore, they are best viewed as system-level failures rather than as failures of an individual or group. The concept of reporting in order to learn and improve is an essential message to disseminate to all personnel. The key to a successful ILS is the analysis and feedback to all staff using the ILS as well as involvement of the staff in the discussion on any subsequent change to practice.

Effective learning from events can be influenced by the skill of the investigators. Investigation of an event requires a broad skill set. It has become clear in Canada, among other countries, that incident investigation and analysis is not always taught in any meaningful way. Typically, an individual, usually a radiation therapist or physicist, is identified to manage the ILS and the event investigation. He/she often has no formal training or experience in event management, and this leads to heterogeneity in practice that is particularly evident at the level of a national ILS. This issue is addressed in key event investigation protocols, like the London Protocol [5], and through short courses or e-learning programmes specific to radiotherapy safety. The Canadian Patients Safety Institute [6] has published a comprehensive event investigation framework that outlines all aspects of event management from assuring that immediate needs of those involved are addressed through investigation, analysis, reporting and change implementation. Canada is also building an online course about event investigation and reporting in radiation oncology to address this issue nationally. The Radiation Oncology Safety Education and Information System (ROEIS) ILS and associated education materials are available to learn about event learning and response [7]. There are also other non-profit organisations, such as the TreatSafely Foundation, that provide both online and in-person educational opportunities in these areas [8].

Maintaining a successful ILS requires several key iterative steps, as shown in Figure 1 (adapted from the Canadian Patient Safety Institute [6]). At a high level, these steps focus on ensuring that the ILS is appropriately supported, events are appropriately addressed and there is effective follow-up, including disseminating lessons learned from event investigations so departments can experience actual improvement in safety.

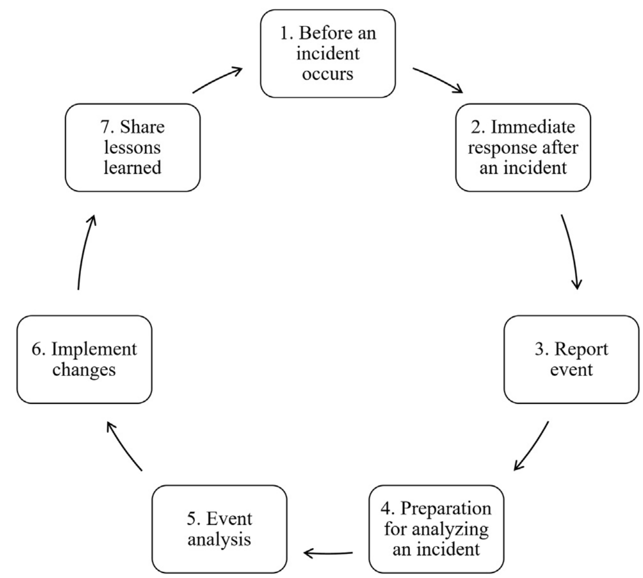

Fig 1. This represents the full spectrum of investigation, analysis, reporting and feedback at the local, national and international levels (adapted from the Canadian Patient Safety Institute [6]). The term "event" refers to either an actual incident or a near-miss.

Key Iterative Steps to a Robust Incident Learning System

(1) Before an event ever occurs, support from leadership is required to cultivate a safe and just culture. A sustainability plan for the ILS needs to be developed that includes resources for on-going support.
(2) Immediately after an event, the patient, family, care provider and anyone else affected must be cared for both emotionally and physically (if necessary). Short-term efforts to reduce the risk of an imminent recurrence should be carried out and any items related to the event should be secured for follow-on analysis. Programme leaders and other key personnel should be notified that an event has occurred, as are regulatory agencies in compliance with local, regional or national requirements.
(3) The results of the event investigation are recorded using a local, national or international ILS.
(4) An investigation team selects an analysis method, sets up meetings, interviews and begins the investigation.
(5) After initial information is collected, the team conducts the analysis, which includes understanding what happened, determining how and why it happened and then develops recommendations to ensure that the probability or severity of future events are reduced.
(6) The recommended remedial actions are acted upon by creating an implementation plan, identifying and mobilising the necessary resources, assigning activities to be carried out and setting deadlines for the activities to be completed. From there, the changes need to be monitored using the ILS to assess their effectiveness.
(7) Finally, it is important to close the loop by sharing what was learned with the patient and his/her family as appropriate, the care team, the radiation treatment programme and other relevant groups or departments both internally and externally.

A worked example of an event report is provided in the Appendix.

Developing Local, National and International Systems

Reporting and learning from events is now a requirement under European legislation, where Article 62 states that Member States shall ensure that quality assurance programmes include a study of the risk of accidental or unintended exposures and the establishment of a system for recording and analysing events or potentially involving accidental or unintended medical exposures. Although ILSs need to capture these sentinel events, it is equally important for systems to be comprehensive and adaptable so as to capture the full spectrum of events along the radiation treatment clinical continuum from the decision to treat until the completion of treatment. Most will be relatively minor deviations in practice with a low likelihood of causing significant dosimetric error or inducing medical harm. Nevertheless, these seemingly insignificant events when considered together can inform system-level change to prevent more severe problems from arising in the future.

ILSs can be local, national or international, each having the same basic goals but facilitating different audiences and environments. A major benefit of the national or international systems is the potential for a larger database of events supporting wider analysis and comparison and sharing of knowledge across a larger community.

Developing an ILS requires the input of all professionals involved in the preparation and delivery of treatment, including radiation oncologists, medical physicists and radiation therapists. This ensures that all perspectives and steps in the process are included.

Local Systems

A local ILS is for the direct benefit of the department in which it is established. Two options are to use the generic hospital incident reporting system if one is already in place or to use a radiation oncology-specific system. Developing and implementing an ILS can be an onerous task and centres often choose to use the generic system, at least in the first instance. The decision should be based on the desired outcomes. Where the ultimate goal is to improve practice, it is important to understand where and how the incident originated and where and by whom (professional discipline rather than an individual) the incident was detected. Given the complexity of the radiotherapy process in terms of both equipment and staff, information of this type is rarely captured in a generic reporting system. Therefore, the potential for true learning towards event reduction is limited. Although it is correct that many hospitals mandate the use of generic ILSs that do not address radiation oncology reporting needs, it may be possible to integrate the radiation oncology taxonomy into the hospital system. This is a long-term aspirational goal, but a few Canadian centres have been able to achieve this and ILS software vendors are becoming increasingly interested.

As part of the learning process it is necessary to identify where in the process it occurred and the safety barriers that were ineffective in preventing it. It is necessary to interrogate all the steps in the process, together with the associated policies and procedures to enable effective change [9]. This level of interrogation cannot be carried out without the detailed description of the event afforded by a dedicated system. Local systems also allow a detailed scrutiny of the data to draw conclusions about patient- and treatment-specific factors that highlight a higher patient risk of harm [10].

It must be borne in mind that local reporting systems are usually grounded in local practice and evolve over time to address specific quality, safety and system performance concerns, many of which are unique to the local environment and not necessarily relevant at national or international levels.

National Systems

A national ILS creates a greater pool to learn from but also requires a standardised approach across the centres in the country. Sharing information across centres provides a platform for the dissemination of information and prevents the unnecessary repetition of errors due to a lack of knowledge of the risks, particularly when a new piece of equipment or a technique is being introduced. However, a national ILS may not satisfy local needs, which can create conflict for individual centres who may want to adopt the national taxonomy and reporting system but do not want to abandon their long-standing in-house or local system. This can lead to concerns about dual reporting and duplicate data entry that translate to slow uptake at the national level. A national ILS can benefit new or small centres that might find adoption of a widely used established system easier than developing their own.

It is important to recognise that the development and implementation of a national ILS requires input from all relevant professional disciplines and all radiation treatment programmes within the country to balance competing priorities, address privacy and confidentiality concerns and, ultimately, assure broad uptake and utilisation. Participation in a national ILS can also help to establish an optimal safety culture at the department level, as has been experienced with the USA-based ILS [11].

International Systems

At an international level, the dissemination of information reaches an even wider audience and has additional benefits, where smaller or developing centres can learn from the experiences of more established centres. An international ILS is in essence altruistic and can increase global awareness of potential issues, identifying system design flaws and safety critical steps in the radiotherapy pathway. It is a reality that incidents do happen; the same incidents occur in different radiotherapy departments and in different countries; information can be shared to anticipate and prevent further incidents.

Based on a study of the literature, Ganesh [12] proposed that 600 events probably occur before any major accident occurrence. This further emphasises the importance of reporting and learning. An international system can potentially provide a larger pool of data, thereby speeding up the collection and analysis process and subsequent learning. Access to international incident data can be very beneficial and a source of learning when new centres are established in countries that have little prior experience with radiation treatment equipment or programme development and operation.

The first international voluntary ILS, Radiation Oncology Safety Information System (ROSIS), was initiated through a European Society of Radiotherapy and Oncology (ESTRO) project under the European Commission Directorate General Health and Consumer Protection: Europe Against Cancer Program in 2001. This system has grown and been transformed into a new system called ROSEIS. The aim was to encourage broader sharing of information about events across a wide range of radiotherapy centres to enable learning and prevent event re-occurrence. ESTRO funded the development of the system over a 2 year period from 2001 to 2003. In 2003, the system went 'live' and was managed from that time forward by a group comprising two physicists and two radiation therapists. The initial system relied heavily on free text descriptions of the events reported. Over time the difficulties in analysing and sharing information in this format became evident and the ILS evolved to incorporate a larger number of defined fields. This was consistent with other local and national systems, which were also moving to taxonomy conformity to increase the learning potential.

The International Atomic Energy Agency (IAEA) also has an ILS tool for radiotherapy patient safety. The IAEA system is called Safety Reporting and Learning System for Radiotherapy (SAFRON) [13] and allows radiotherapy centres internationally to contribute events for the purpose of learning. SAFRON and ROSEIS have been designed to be both a local and an international incident reporting system to provide the maximum benefit to individual centres and the wider community. The ROSEIS and SAFRON taxonomies are broadly compatible, which will facilitate data sharing or comparative analysis.

Operating Local, National and International Systems

A key component of any ILS is timely and effective analysis of events, as well as feedback to the users of the system. It is important for staff to know that reports are acknowledged, analysed and acted upon.

Local Systems

Local reporting is only one part of the event investigation and management spectrum (Figure 1). To be effective, event reporting needs to be embedded in a programmatic approach that includes top-down support from leadership, expert oversight in the form of an interdisciplinary quality committee and policies and procedures that define how reports are submitted, reviewed and acted upon. For example, a multidisciplinary team could meet weekly [14]. Guidelines are available to help local programmes operationalise incident management. These include the Canadian suite of radiation treatment programme key quality indicators, which have now been incorporated as an important part of cancer programme accreditation across the country. A review of other relevant local systems can provide the background information on which to build, thus minimising duplication of effort. The taxonomy should be carefully chosen to avoid having to redesign the system should consideration be given to wider sharing and learning at national or international level in the future. The environment must be considered and a change of culture may well be necessary. A hierarchical management approach to reporting and learning will not succeed unless all personnel involved in the delivery of radiotherapy are involved and have ownership of the final system.

National Systems

National ILSs that aggregate event data from multiple centres are valuable because they increase the number of events and the overall power of learning but also bring challenges associated with utilisation, confidentiality and privacy, and data quality, analysis and ownership. Except for mandatory reporting to regulatory agencies of radiation overdoses affecting patients, the public or health care providers, most national reporting is voluntary and at the discretion of individual radiation treatment programmes. A combination of top-down and bottom-up strategies is necessary to assure optimal performance of a national system. On the one hand, national, regional and local cancer programme leadership needs to acknowledge the value of event reporting and learning, assure that resources are appropriately allocated for initial development and ongoing operation of the system and help break down barriers associated with the sharing and aggregation of data across institutions. Initial grass-roots efforts to build broad consensus about the objectives and scope of a national system, develop a taxonomy that is relevant and generally compatible with existing local systems and streamline the reporting process to minimise duplication of work will go a long way to assure utilisation [15].

Many programmes will not want to abandon their long-standing local systems. Therefore, anything that reduces the burden of data entry will probably be helpful, such as batch electronic uploading of data entered locally to the national system. In addition, all invested in the development and use of the system need to see value for their efforts, which might include real-time alerts about serious events with the potential to re-occur in other centres or regular reports about key themes that can be used to inform process change. This requires a national oversight group with a mandate and the necessary expertise to monitor data quality, review and analyse events, identify patterns and trends and assure timely reporting back to the community.

International Systems

With an international system, anonymity is crucial and must be maintained if such a system is to succeed. Both current international systems, ROSEIS and SAFRON, support anonymous reporting. Access to both systems is through the organisation websites but is restricted to registered users.

Although an international system brings substantial and positive benefits to the wider radiotherapy community, it must be embedded within an organisation that can support it in the long term, as the operation of an international system presents significant challenges. The original ROSIS system [16] was altruistic in nature but the lack of dedicated funding did affect its sustainability. The support of radiation oncology vendors such as Elekta (www.elekta.com) and Varian (www.varian.com) to further develop the system was of great benefit and facilitated the revision and subsequent absorption into the ESTRO platform.

The development of SAFRON within the IAEA and the integration of ROSEIS within ESTRO give stability and security to the systems. The SAFRON system is managed by the Division of Nuclear Safety & Security of the IAEA and ROSEIS has recently been integrated into the ESTRO IT infrastructure and will be overseen by ESTRO's Radiation Oncology Safety and Quality Committee with respect to content development and management. This harmonisation recognises that the value of international systems is in sharing information about key themes across jurisdictions.

Choosing an Incident Learning System Structure and Taxonomy

Harmonisation of reporting systems at local, national and international levels is key to maximising the learning potential. Individual programmes should adopt one of the national or international radiation treatment taxonomies that is most relevant to their practice.

A taxonomy is simply a classification scheme to organise concepts into a unifying framework for discussion and data analysis. The value of the taxonomy depends on many factors. For example, the taxonomy must be intuitive and easy to use. If clinicians spend too much time trying to remember the classification of the taxonomy, then it will not get used. The taxonomy should be robust so that multiple clinicians trying to classify the same event naturally use the same descriptors. Finally, the taxonomy should be capable of being mapped to taxonomies from other ILSs, whether local, national or international.

A structure for ILSs in radiation oncology has been proposed and includes a causality table [17]. The point of the causality table is to create consistency in identifying the contributing factors to an event and then being able to share that information with others. More broadly, for example, the Canadian taxonomy system included six data groups of: impact, discovery, patient, details, treatment delivery and investigation [15]. This taxonomy has been recently updated and now includes 35 items in the six groups. An example of this taxonomy structure and data groups with one item for each group is shown in Table 1. Reporting systems at the local and national levels should be harmonised and this requires a robust taxonomy.

Table 1
Abbreviated example of a taxonomy structure adapted from the revised (2017) Canadian national system for incident reporting in radiation treatment (NSIR-RT) taxonomy

Data group	Item in data group	List of all choices in this data group item
Impact	Type of radiation treatment	Actual incident; Near-miss; Programmatic hazard
Discovery	Health care provider(s) or other individuals involved in the event	Radiation therapist; Radiation therapy team supervisor; Treatment planner; Medical physicist; Radiation oncologist; Nursing staff; Resident or fellow; Administrator; Patient; Family member; Other
Patient	Patient gender	Male; Female; Undifferentiated; Do not know
Details	Problem type	Inappropriate or poorly informed decision to treat or plan; Wrong side (laterality); Wrong prescription dose-fractionation or calculation error; Failure to perform on-treatment imaging as per instructions; Wrong patient; Wrong patient position set-up point or shift; Wrong target or organ at risk contours; Treatment plan (isodose distribution) unacceptable; Other
Treatment deliver	Body region(s) treated	Abdomen; Brain; Breast; Thorax; Head and neck; Pelvis; Upper extremity; Lower extremity; Spine; Skin
Investigation	Safety barrier(s) that failed to identify the incident	Hardware/software process: Use of record and verify system; Image-based patient position verification; Interlocks; Regular equipment performance verification; Regular external dosimetry and performance audit Process: Verification of patient ID; Verification of relevant clinical information; Oncologist peer review; Time out/verbalisation/call-back; Verification of treatment accessories; Therapist review of treatment position; On-treatment therapist chart check; Regular chart check

Conclusions

Overall there is a clear need for an ILS at local, national and international levels supporting incident reduction and increasing quality and safety across the entire radiotherapy community. Furthermore, they have been shown to be effective in improving safety [18]. Reporting and learning from events is a key component of quality and safety in radiotherapy.

Appendix: Worked Example

Vignette

An experienced therapist (Jack) is working at one of the linear accelerators and is getting ready to treat Mrs Smith. A junior therapist from computed tomography simulation (Jill) has joined Jack today because of short staffing and two therapists are required to work together on the linear accelerators at all times. Mrs Smith is receiving conventional course three-dimensional radiotherapy for lung cancer. Today, Jack and Jill are getting ready to treat the second fraction of the boost (phase 2), which is planned for 1080 cGy in six fractions. The phase 1 prescription was 4500 cGy in 25 fractions. Jack acquired orthogonal kV images per the prescription that were used to set up Mrs Smith for treatment. The set-up and treatment went smoothly, with Jill performing a review of the images and time-out before the treatment. Later, Jack felt uneasy about Mrs Smith's treatment. When he reviewed the images, he realised that Mrs Smith was treated one vertebral body superior from the planned isocentre.

Event Report

Patient XXX was setup and treatment error 1 vertebral body superior to the plan on the 16th fraction of treatment. Conventionally fractionated lung treatment and it was the 2nd fraction of the boost (1080 cGy/fraction · 6 fractions). The CT simulation therapist was helping cover the linac because the senior therapist initially working alone. It was a routine busy day at the linac and this patient tended to set up a bit differently every day. The senior therapist felt uneasy and went back to check on the patient setup images and detected the setup error. Classification of the event following a taxonomy (See Table 1)

Data group	Item within data group
Impact	Actual incident
Discovery	Radiation therapist
Patient	Female
Details	Wrong patient position, set-up point or shift
Treatment deliver	Thorax
Investigation	Hardware/software process: image-based patient position verification Process: therapist review of treatment position

Note: A video of this incident can be found at https://i.treatsafely.org/search-view-module/118350878 and associated paperwork can be found at https://i.treatsafely.org/get-gs-file/38628. These are free resources that can be used to facilitate learning within your department.

References

[1] Leunens G, Verstraete J, Van den Bogaert W, *et al*. Human errors in data transfer during the preparation and delivery of radiation treatment affecting the final result "garbage in, garbage out". *Radiother Oncol* 1992;23(4):217–222.
[2] Shafiq J, Barton M, Noble D, *et al*. An international review of patient safety measures in radiotherapy practice. *Radiother Oncol* 2009;92(1):15–21.
[3] Institute of Medicine (IOM) report. *Patient safety: achieving a new standard for care*. Washington, D.C., USA: National Academy Press; 2004.
[4] Council Directive 2013/59/EURATOM. *Laying down basic safety standards for protection against the danger arising from exposure to ionizing radiation*. Official Journal of the European Union; 5 December 2013.
[5] Taylor-Adams S, Vincent C. Systems analysis of clinical incidents: the London Protocol. *Clinical Risk* 2004;10:211–220.
[6] Canadian Patient Safety Institute (CPSI). Available at: http://www.patientsafetyinstitute.ca (accessed 16 May 2017).
[7] Radiation Oncology Safety Education Information System (ROSEIS). Available at: https://roseis.estro.org (accessed 16 May 2017).
[8] TreatSafely Foundation. Available at: https://www.treatsafelyfoundation.org (accessed 16 May 2017).
[9] Marks LB, Jackson M, Xie L, *et al*. The challenge of maximizing safety in radiation oncology. *Pract Radiat Oncol* 2011;1(1):2–14.
[10] Elnahal SM, Blackford A, Smith K, *et al*. Identifying predictive factors for reports in patients receiving radiation therapy. *Int J Radiat Oncol Biol Phys* 2014;90(5):1202–1207.
[11] Hoopes DJ, Dicker AP, Eads NL, *et al*. RO-ILS: Radiation Oncology Incident Learning System: a report from the first year of experience. *Pract Radiat Oncol* 2015;5(5):312–318.
[12] Ganesh T. Incident reporting and learning in radiation oncology: need of the hour. *J Med Phys* 2014;4:203–205.
[13] Holmberg O, Malone J, Rehani M, *et al*. Current issues and actions in radiation protection of patients. *Eur J Radiol* 2010;76:15–19.
[14] Rahn DA, Kim GY, Mundt AJ, *et al*. A real-time safety and quality reporting system: assessment of clinical data and staff participation. *Int J Radiat Oncol Biol Phys* 2014;90(5):1202–1207.
[15] Milosevic M, Angers C, Liszewski B, *et al*. The Canadian National System for Incident Reporting in Radiation Treatment (NSIR-RT) Taxonomy. *Pract Radiat Oncol* 2016;6(5):334–341.
[16] Cunningham J, Coffey M, Knöös T, *et al*. Radiation Oncology Safety Information System (ROSIS) – profiles of participants and the first 1074 incident reports. *Radiother Oncol* 2010;3:601–607.
[17] Ford EC, Fong de Los Santos L, Pawlicki T, *et al*. Consensus recommendations for incident learning database structures in radiation oncology. *Med Phys* 2012;9(12):7272–7290.
[18] Nyflot MJ, Zeng J, Kusano AS, *et al*. Metrics of success: measuring impact of a department near-miss incident learning system. *Pract Radiat Oncol* 2015;5(5):334–341.